

A 2.9–4.0-GHz Fractional-N Digital PLL With Bang-Bang Phase Detector and 560-fs_{rms} Integrated Jitter at 4.5-mW Power

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Abstract—This paper introduces a $\Delta\Sigma$ fractional-N digital PLL based on a single-bit TDC. A digital-to-time converter, placed in the feedback path, cancels out the quantization noise introduced by the dithering of the frequency divider modulus and permits to achieve low noise at low power. The PLL is implemented in a standard 65-nm CMOS process. It achieves -102 -dBc/Hz phase noise at 50-kHz offset and a total absolute jitter below 560 fs_{rms} (integrated from 3 kHz to 30 MHz), even in the worst-case of a -42 -dBc in-band fractional spur. The synthesizer tuning range spans from 2.92 GHz to 4.05 GHz with 70-Hz resolution. The total power consumption is 4.5 mW, which leads to the best jitter-power trade-off obtained with a fractional-N synthesizer. The synthesizer demonstrates the capability of frequency modulation up to 1.25-Mb/s data rate.

Index Terms—ADPLL, bang-bang, DCO, DPLL, fractional-N, frequency synthesis, jitter, lead-lag, phase noise, spur, TDC, TDC-less.

I. INTRODUCTION

THE advent of ultra-scaled CMOS technologies is leading to the digital implementation of critical subsystems, which traditionally belonged to the realm of analog circuits. This process is sustained by CMOS device scaling which offers both high-performance digital signal processing at very low cost and better device matching for non-minimum-size devices. As an example, the trade-offs among resolution, conversion rate and power dissipation in analog-to-digital [1] and digital-to-analog converters (ADCs and DACs) improve with device scaling. Digitization not only entails the clear advantages of repeatability and portability to new scaled processes, but it also paves the way to the extensive application of powerful techniques for correcting the variable analog impairments.

The frequency synthesizer, conventionally used for coherent demodulation/modulation in wireless transceivers, is one of the building blocks most involved in this evolution. A wealth of

literature has been devoted to this topic, since the first digital phase-locked loop (PLL) for those applications was presented in [2], less than one decade ago. The interest in this class of circuits is expected to increase with the introduction of fourth-generation high-data-rate communication standards, in particular LTE or WiMAX, that require wideband radios with high signal-to-noise ratio and may demand a local oscillator with integral phase noise lower than -35 dB. This requirement is sometimes expressed by means of the absolute jitter that is, roughly speaking, the RMS value of the difference between the zero-crossing instants of the PLL output with respect to an ideal noiseless clock.¹ As an example, at 3.6 GHz the specification of integral noise less than -35 dB translates into a jitter lower than 790 fs_{rms}. This stringent requirement poses limitations to both tolerable phase noise and spurs level. The improvement of noise and spurs performance always costs an increase in power dissipation; therefore, an important challenge is designing a high-efficiency frequency synthesizer, which in other words has low jitter at low power. A figure-of-merit (FoM), which can characterize a frequency synthesizer in terms of this jitter-power trade-off, has been introduced in [3], as the product of the jitter variance σ_t^2 and the power consumption P expressed in milliwatts.

$$\text{FoM} = 10 \cdot \log_{10} \left[\left(\frac{\sigma_t}{1 \text{ sec}} \right)^2 \cdot \left(\frac{P}{1 \text{ mW}} \right) \right].$$

State-of-the-art analog and digital fractional-N synthesizers are limited to a FoM of about -233 dB. This value becomes higher when in-band fractional spurs are taken into account. By comparing those values to the best FoM of -251 dB obtained in [3] for an integer-N synthesizer, we can realize how large is the efficiency gap still existing between integer- and fractional-N synthesizers. The main goal of the present work is to reduce this gap and to design a fractional-N digital synthesizer with a FoM closer to the one of the best integer-N synthesizer, but with finer frequency resolution (below 100 Hz) and with wideband frequency modulation capability. The latter feature is essential for transmitters employing direct frequency modulation such as the polar architecture.

A digital PLL can be directly derived from the $\Delta\Sigma$ fractional-N analog PLL and the resulting simplified schematic, sometimes referred as $\Delta\Sigma$ fractional-N digital PLL, is shown

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¹This figure should not be confused with the period (or cycle) jitter, which measures the RMS value of the stochastic process $T[k]$ representing the error between the current period and the period of an ideal clock. If $t[k]$ is the difference between the real and ideal k th zero-crossing instant, it is $T[k] = t[k] - t[k-1]$.

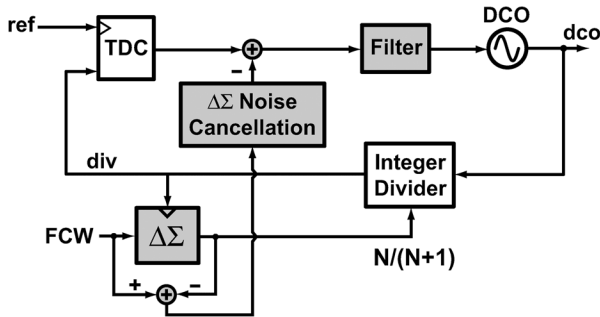


Fig. 1. A digital fractional-N PLL employing a multi-bit high-resolution TDC.

in Fig. 1. In this topology, the $\Delta\Sigma$ quantization noise is subtracted from the output of the time-to-digital converter (TDC), acting as a phase detector (PD) [4]–[6]. This cancellation technique has been also applied to analog PLLs, but the digital version allows easy and accurate implementation of this algorithm. The jitter-power trade-off, in this circuit, is mainly set by the TDC, which is analogous to an ADC. The TDC finite resolution may represent the ultimate limit to the PLL in-band noise, but more often, it causes concentrated fractional spurs [7]. As an example, a TDC having 1-ps resolution and employed in a 3.6-GHz PLL with a 40-MHz reference clock should produce a -120 -dB/Hz in-band noise at the output, under the assumption of uniform distribution of the TDC quantization noise. In fact, this figure has not been achieved in digital PLLs yet. The problem is that the finite resolution of TDCs gives rise to spurious tones in PLLs. Such tones typically fall within the PLL bandwidth, when a near-integer channel must be synthesized, and substantially increase the output jitter. This justifies the current research effort in realizing TDCs with sub-picosecond resolution for these applications.

In addition to resolution, the design of TDCs for digital PLLs must focus on the linearity of its characteristic, since nonlinearity is another source of fractional spur generation. All TDCs based on flash conversion trades resolution and linearity against power dissipation. The companion digital topology typically referred as “phase-domain” all-digital PLL (ADPLL) [2] presents similar issues and the same jitter-power trade-off.

Several attempts have been done in recent literature to increase TDC resolution at low power. Improvements have been obtained by employing the subranging approach as in [8] and in [9] or by exploiting the metastability phenomenon for time amplification [10], [11]. Another interesting approach is the gated-ring oscillator (GRO) [12], which relaxes the resolution-power trade-off, taking advantage of the accumulation of quantization error, typical of $\Delta\Sigma$ converters. However, this approach complicates the design and makes performance sensitive to voltage supply. More recently, Vercesi *et al.* have proposed a two-dimensional Vernier-type TDC, which minimizes the number of delay elements required, thus improving efficiency [13], but at the price of a more complex design. Other works have faced the other important issue in TDCs, i.e., the linearity, by implementing a sliding-scale technique [14] or, similarly, by shuffling randomly the TDC elements [6]. In practice, all high-performance TDCs dissipate a large part of the whole power budget,

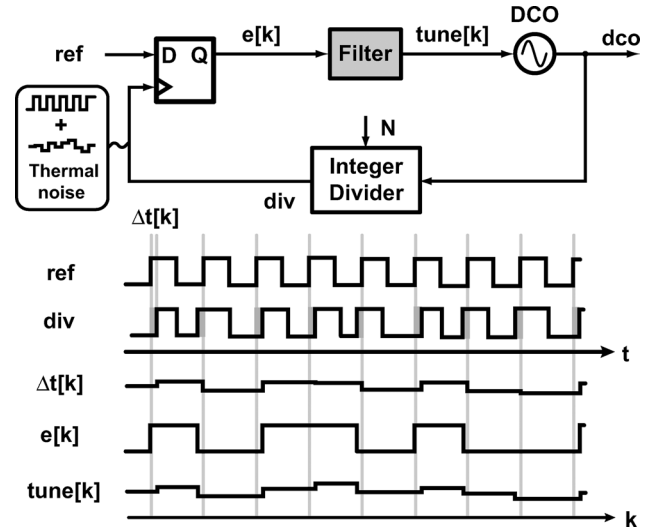


Fig. 2. Waveforms in an integer-N bang-bang PLL.

worsening the jitter-power trade-off and complicating greatly the synthesizer design.

In this paper, we propose a different approach, which consists in adopting a single-bit TDC as phase detector [15]. With respect to the standard structure of a digital PLL, a high-resolution controllable delay, i.e. a digital-to-time converter (DTC), is used in place of a high-resolution TDC. This choice leads to easier design and better jitter-power compromise. Compared to a TDC based on a delay line with flip-flops, a DTC is obtained by using a delay line or a delay-varying element. The elimination of flip-flops leads to lower power consumption and lower nonlinearity, since the latter is mainly caused by the mismatches in the delay elements and not by the additional mismatches contributed by the flip-flops.

Single-bit (also called bang-bang or lead-lag) PDs are employed in specific applications of PLLs, such as clock and data recovery (CDR), and they have recently drawn new interest in the field of frequency synthesis, e.g. [16], [17]. Methods for extending the use of bang-bang control to the fractional-N case have been shown in [18] and [19]. However, as it will be clearer in the following, those methods are not applicable to high-performance and high-efficiency frequency synthesizers, needed in wireless applications.

The paper is organized as follows: The next section discusses the problems that hinder the realization of a bang-bang PLL for wireless standards requiring low integrated phase noise (such as WiMAX). This analysis suggests the solution, presented in Section III, based on the use of a true fractional-N divider in the feedback path. Section IV discusses the design of this block, while Section V briefly considers the other building blocks of the entire fractional-N bang-bang PLL. The measurements are presented in Section VI and, finally, the conclusions are drawn in Section VII.

II. ISSUES IN THE REALIZATION OF A FRACTIONAL-N BANG-BANG PLL

A single-bit TDC, which will be referred in the following as bang-bang PD (BB-PD), consists of a single D-flip-flop

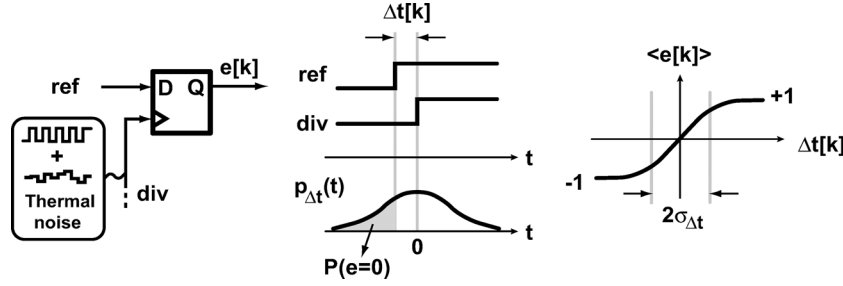


Fig. 3. Linear model for the BB-PD.

detecting which one of its two input signals (the data and the clock) leads or lags the other one. Bang-bang PDs therefore act as hard limiters on the time error between the rising edges of the reference and the divider output. This nonlinearity can be responsible of huge undesirable spurs and noise at the PLL output, especially when a fractional channel is synthesized. The situation may be instead different under specific conditions, when an integer division is performed. We consider first the latter situation (shown in Fig. 2), since the understanding of the integer-N case is useful for the illustration of the fractional-N one.

Intuitively, when the bang-bang PLL (BB-PLL) is locked, the rising edges of the divider output (div) wander around the rising edges of the reference signal (ref). For instance, if the k th rising edge of ref (ref[k]) leads div[k], the time error $\Delta t[k]$ will be larger than zero and the TDC output will be $e[k] = 1$. Since in locked condition the loop filter can be considered a simple gain,² β , the tuning of the digitally-controlled oscillator (DCO) increments by β and its output period is decreased by βK_T (where K_T is the period gain of the DCO). The period of the divider output is therefore shortened by an amount $N\beta K_T$ (where N is the division factor), reducing the delay $\Delta t[k + 1]$ (even to negative values) with respect to $\Delta t[k]$. In absence of any noise source, this behavior would give rise to a limit cycle or, equivalently, to a modulation of the output frequency, which is unacceptable in a synthesizer for wireless applications. Essentially, this limit cycle is caused by the effect of the “frequency granularity” in the DCO, discussed by several authors. However, if the RMS time jitter referred to the BB-PD input, $\sigma_{\Delta t}$, caused by all the unavoidable physical noise sources within the PLL, is of the same order of $N\beta K_T$ or larger than that, the limit cycle will be dithered by the noise itself and the output spurs will be eliminated. A rigorous analysis of this behavior can be found in [17]. This mode of operation can be referred as “random-noise regime”, as discussed in [20], and it is schematically shown in Fig. 2, where the signals do not display any periodicity.

Two additional remarks may be helpful. First, in presence of noise dithering the limit cycle, it is possible to derive a linearized model of the whole loop, as described in [21]. According to that work, the BB-PD can be represented as a constant gain, plus some quantization noise. That model can be intuitively justified as follows: We can consider, for instance, the case in which

²The loop filter is usually a proportional-integral (PI) filter. However, the coefficient of the integral path is much smaller than the one of the proportional path. Thus, it can be neglected in a first-order analysis when the PLL is locked [17], [19].

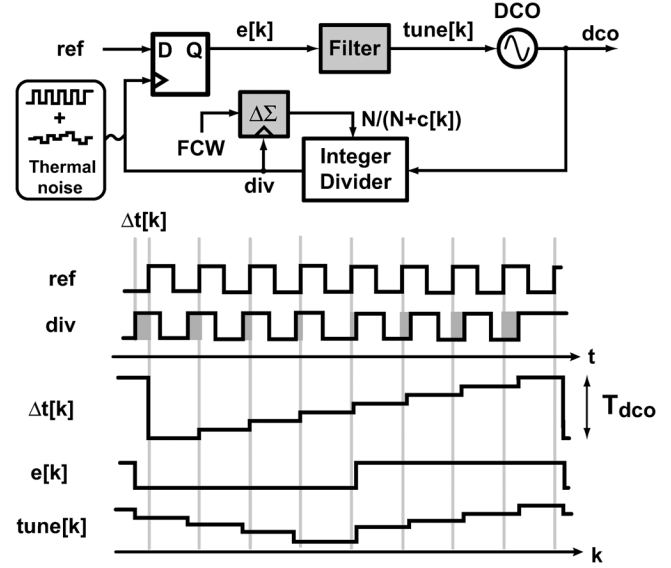


Fig. 4. Waveforms in a fractional-N bang-bang PLL.

$\Delta t[k]$ is larger than zero, as in Fig. 3. Although the output $e[k]$ should be one, the jitter produced by the unavoidable random noise can cause $e[k]$ to be equal to zero. In Fig. 3, we have assumed that the entire jitter is equivalently superimposed to the edge of the div signal and that it has a Gaussian distribution. Since the PLL bandwidth is much lower than $1/T_{\text{ref}}$ (being T_{ref} the reference period) what counts for the loop behavior is the characteristic that links $\Delta t[k]$ and the time average of $e[k]$. From Fig. 3, it results that this average, $\langle e[k] \rangle$, is proportional to the integral of the Gaussian density from $-\Delta t[k]$ to positive infinity. The resulting characteristic of the BB-PD, see Fig. 3, can be approximated with a linear gain K_{bpd} around $\Delta t[k] = 0$. It is intuitive that this gain is proportional to $1/\sigma_{\Delta t}$. The rigorous analysis in [21] gives $K_{\text{bpd}} = \sqrt{2}/(\sigma_{\Delta t}\sqrt{\pi})$ and defines a linear input range of about $2\sigma_{\Delta t}$. As a second remark, note that the loop in Fig. 2 reduces to a Δ modulator, whose input is the analog signal consisting in the zero crossing instants of the input ref. In fact, if the filter is assumed a simple gain block, the circuit is simply an integrator (i.e. the DCO) closed in a feedback loop by the hard-limiter (i.e. the flip-flop).

The previous considerations about the integer-N PLL help in understanding the problems in realizing a bang-bang fractional-N PLL. Fig. 4 shows the case in which a 1st-order $\Delta\Sigma$ modulator (a simple accumulator whose carry is the modulator output) dithers the division factor. It is well known that the

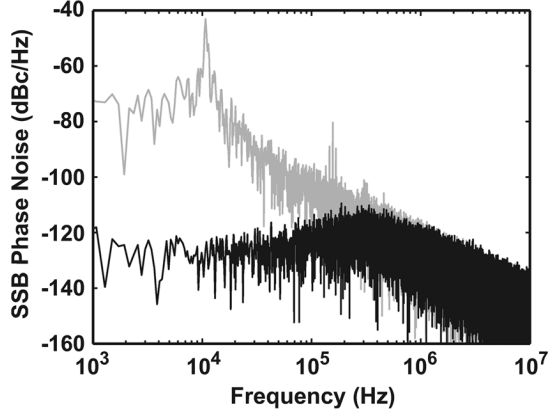


Fig. 5. Simulated spectra of the bang-bang PLL when an integer channel (black line) and a fractional channel (gray line) are synthesized, respectively.

steady-state time difference $\Delta t[k]$ between ref and div increases as a ramp. Then, it experiences a large step of amplitude T_{dco} (the PLL output period), when the carry output changes to one, resulting in a periodic sawtooth-like signal. The frequency of this waveform is equal to the fractional part of the frequency control word (FCW) times the reference frequency f_{ref} . The peak-to-peak amplitude, T_{dco} , of this deterministic signal is much larger than the random component of the time jitter $\sigma_{\Delta t}$. As an example, T_{dco} is in the range of 250–300 ps for an output frequency between 3 and 4 GHz, while $\sigma_{\Delta t}$ may be estimated in the order of hundreds of femtoseconds for high-performance applications, thus roughly a factor 10^3 below. In this way, the dithering effect of the random noise is, in practice, completely lost. Fig. 4 shows that the TDC output, $e[k]$, in this condition is a square wave with the same period of the $\Delta t[k]$ sawtooth signal. The tune[k] tuning signal is not constant. In fact, the integral path of the PI loop filter in this case cannot be neglected. When $e[k]$ is constant, the amplitude of each step of tune[k] is equal to the coefficient of the integral path. The presence of a limit cycle is evident: The periodic behavior of the tuning voltage will give rise to large fractional spurs which may fall within the PLL bandwidth or, in general, to a serious degradation of the output spectrum, because of the strong noise folding due to the presence of the hard nonlinearity, an effect similar to spectral regrowth. To support this conclusion, it is possible to simulate the bang-bang PLL, both when the division factor is integer and when a fractional division factor is obtained by the $\Delta\Sigma$ dithering of the frequency divider modulus. Fig. 5 shows the simulation results where, as expected, the spectrum in the fractional case (gray line) exhibits much larger noise and spurs than the spectrum in the integer case (black line). In the example in Fig. 4, we have considered a 1st-order $\Delta\Sigma$ modulator, for the sake of clarity. However, the same problem, evidenced in that figure, also occurs when a higher-order modulator is employed. In fact, the advantages derived from the much more “randomized” output coming from the higher-order $\Delta\Sigma$ modulator are nullified by a much larger peak-to-peak time error. The synthesizer presented in [18] faces the issue of the PD hard nonlinearity by closing it in a loop with the $\Delta\Sigma$ modulator. However, this results into a type-I PLL, with intrinsically poorer phase noise performance.

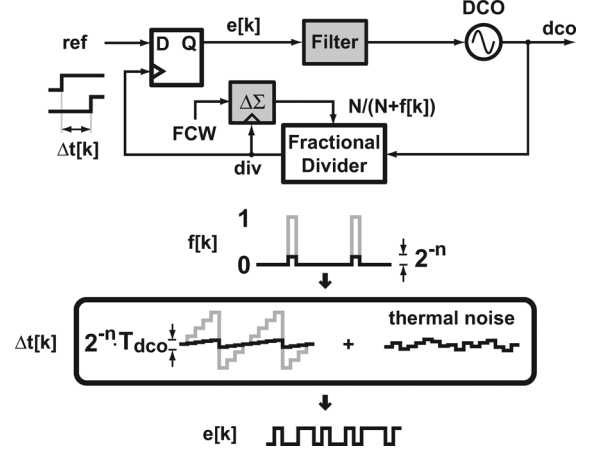


Fig. 6. Recovering the random noise regime when a true fractional divider is employed.

The adoption of a BB-PD in a fractional-N PLL entails other issues, in addition to the one discussed above: (i) The coarse quantization of the time error does not allow the cancellation of the $\Delta\Sigma$ noise at the TDC output (as it was instead possible in the scheme in Fig. 1). (ii) When the PLL operates in the thermal-noise regime, the BB-PD behaves as a linear block whose gain depends on the input time jitter $\sigma_{\Delta t}$. Therefore, some PLL characteristics, such as bandwidth, would depend on the amount of PLL noise, which is highly undesirable. (iii) In the random-noise regime, the limited input linear range of the PD could considerably slow down the PLL frequency lock transient.

III. RECOVERING RANDOM NOISE REGIME WITH A TRUE FRACTIONAL DIVIDER

A. Recovering the Random Noise Regime

The above discussion about the bang-bang fractional-N PLL evidences that the main problem of this loop is the large quantization step experienced by the zero-crossing edges of the div signal, which is introduced by the dithering action of the $\Delta\Sigma$ modulator on the divider modulus. The minimum variation of the division factor is one, which induces a time variation equal to T_{dco} , which is much larger than the RMS input jitter.

A possible solution is therefore to substitute the standard integer-N divider with a “true” fractional-N divider, i.e. a divider that is able to vary somehow the period of the div signal by a fraction of T_{dco} . According to the considerations carried out in the previous section, the time resolution of this true fractional divider should be in the order of the RMS input time jitter, roughly $10^{-3}T_{dco}$. Thus, it must be $(2^{-n})T_{dco}$ with $n = 10$. In this fashion, see Fig. 6, the $\Delta\Sigma$ modulator will interpolate the division factor between N and $N + f[k]$, where $f[k]$ is now a fractional number, let us say 2^{-n} . The maximum amplitude of the sequence $\Delta t[k]$ is now limited to $(2^{-n})T_{dco}$ and the sequence values is determined by the random input jitter $\sigma_{\Delta t}$, thus recovering the random-noise regime and eliminating the limit cycle.

It is worth noting that with this solution no external, pseudo-random, dither sequence is added in the loop, to scramble the spurs in contrast to other published works, for instance [16]

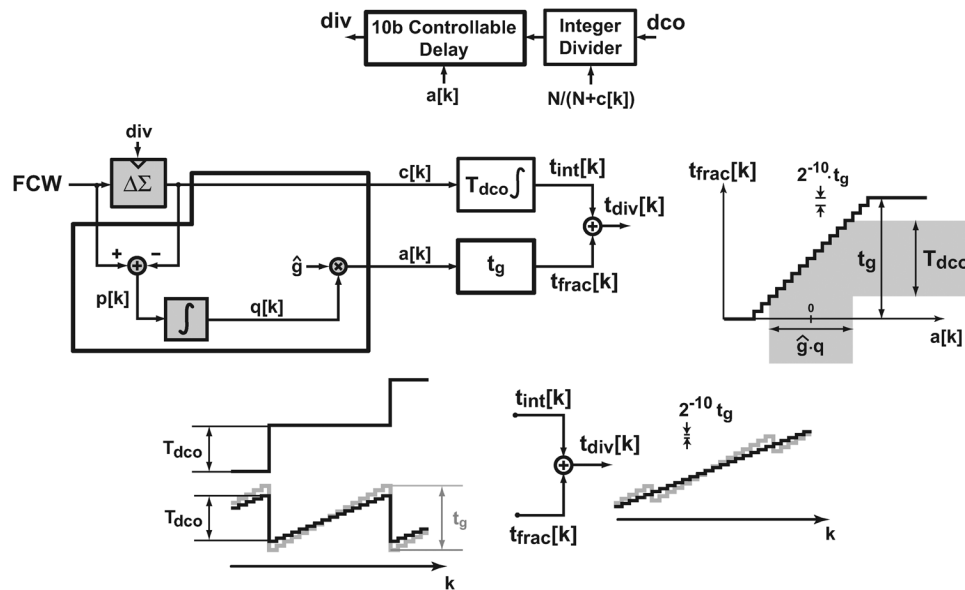


Fig. 7. The proposed fractional divider (top), its time-domain model and the characteristic $a[k]$ vs. $t_{\text{frac}}[k]$ (center), and (bottom) the time diagrams of the excess time delays with $\hat{g} = 1$ (light gray) and when $\hat{g}$ is tuned (black).

and [22]. Those approaches, in fact, are effective in randomizing the spurious tones, but at the price of an increase of the PLL phase noise. This work, instead, exploits the true random physical noise sources (mainly thermal noise), which are unavoidable in a circuit, to eliminate fractional periodicity without adding any noise penalty to the output spectrum. The idea that the inevitable presence of physical noise can be useful in presence of nonlinearities is known in some scientific disciplines as “stochastic resonance”. A very recent account of this topic from the point of view of electronic design can be found in [23].

B. Operating Principle of the Fractional-N Divider

The required fractional-N divider, in principle, could be realized as a phase-interpolator circuit, as in [6]. Unfortunately, in this specific case more than 1 000 phases should be generated, a fact that precludes the use of the interpolator as well as the realization of a multiphase oscillator. In reality, however, only one phase is required at a certain clock cycle, to be compared with the reference signal. Starting from this consideration, we see that the 10-bit fractional-N divider can be realized as an integer-N divider, followed by a digitally-controlled delay stage, as depicted in Fig. 7 (top). The latter circuit, whose design details are given in the following section, is controlled by $a[k]$ and generates the required phase with the desired delay accuracy: It is essentially a digital-to-time converter, the equivalent of a DAC.

The time-domain model, also in Fig. 7, may help to clarify how $a[k]$ is obtained, and in general the operating principle of this circuit. For a better understanding, first note that in a *standard* fractional-N PLL (i.e. without the controllable delay) only the upper path of the model is present, thus it would be $t_{\text{int}}[k] = t_{\text{div}}[k]$. The $t_{\text{div}}[k]$ signal represents the excess time delay at the divider output, i.e. the additional delay with respect to the case in which the division factor is integer. In the ideal case of a true fractional-N divider, this signal should be a

linear ramp. Instead, in a standard fractional-N PLL driven for simplicity by a 1st-order $\Delta\Sigma$ modulator, $t_{\text{div}}[k]$ would be a staircase-like signal (with steps of T_{dco} amplitude occurring when $c[k] = 1$) and would produce the sawtooth-like error $\Delta t[k]$ shown in Fig. 4. Accordingly, the divider is modeled as an integrator, multiplied by a gain T_{dco} (See Fig. 7).

In the proposed circuit an additional time shift, $t_{\text{frac}}[k]$, is added by the 10-bit DTC, to reduce the error $\Delta t[k]$. This extra delay can be represented by a second parallel path, given that now it is $t_{\text{div}}[k] = t_{\text{int}}[k] + t_{\text{frac}}[k]$. Since the parallel path must reduce the large quantization error due to the dithering between N and $(N + 1)$, it is reasonable that $a[k]$ is obtained by first extracting the $\Delta\Sigma$ quantization noise, $p[k]$, then by integrating $p[k]$ to obtain $q[k]$. This integration, which matches the integration in the other path, is necessary to convert the variable “frequency” (controlled by the $\Delta\Sigma$ modulator) into the variable “time delay”.

The $q[k]$ signal is then corrected by a proper gain $\hat{g}$ and finally converted into a time delay by the 10-bit DTC. The static characteristic of this quantizer, $t_{\text{frac}}[k]$ versus $a[k]$, is also shown in Fig. 7, where its full-scale range is indicated as t_g (i.e. when $a[k]$ is one, $t_{\text{frac}}[k]$ increments by t_g). By design, it is $t_g > T_{\text{dco}}$. Thus, by adjusting $\hat{g}$, it is possible to adapt the DTC input range $\hat{g} \cdot q$, so that its output range equals the DCO period T_{dco} . In this way, the two paths perform the same function, since they are both integrators with gain T_{dco} . The only difference is that the second one has a much finer resolution. In fact, the $t_{\text{int}}[k]$ minimum increment is T_{dco} , while the minimum time step induced by the controllable delay $t_{\text{frac}}[k]$ is $2^{-10}t_g$, i.e. about $2^{-10}T_{\text{dco}}$.

The need for an adjustable gain $\hat{g}$ (and consequently, the necessity of having $t_g > T_{dco}$) is dictated by the practical impossibility to obtain exactly $t_g = T_{dco}$, because of the process, voltage and temperature (PVT) spreads. Fig. 7 (bottom) also shows the temporal evolution (as k increases) of the output excess time shift by using the time-domain model, in two different

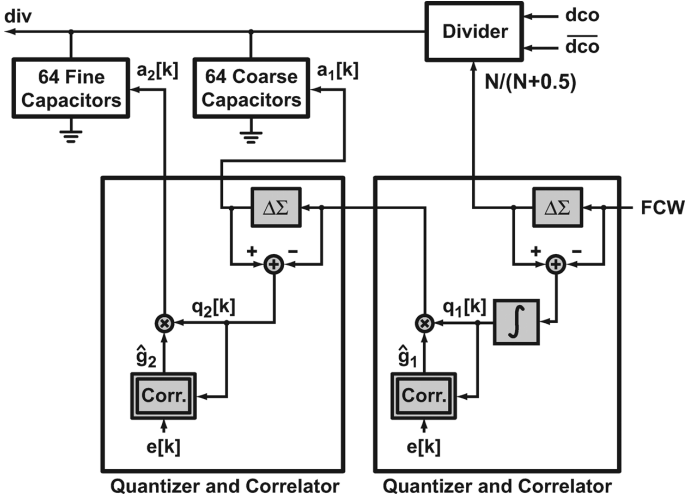


Fig. 8. Block schematic of the true fractional divider (the DTC is implemented with a segmented architecture).

cases. In the first one (light-gray diagram), where $\hat{g}$ is equal to 1, the $t_{frac}[k]$ sawtooth signal features an amplitude $t_g > T_{dco}$. Thus, after the summation of the two delays, the unwrapped ramp displays some periodic modulation, which translates into unwanted fractional spurs. When $\hat{g}$ is regulated so that the DTC output full-scale is reduced to T_{dco} (black diagram), the unwrapped ramp shows no error.

The gain $\hat{g}$ can be regulated in background by adopting the digital techniques well known in the field of adaptive filters. In principle, by correlating the $q[k]$ signal with $e[k]$, the 1-bit output of the BB-PD, the desired value of the gain $\hat{g}$ is obtained. In practice, this solution implements a sign-least mean squares (sign-LMS) algorithm [24].

Although the operating principle of the divider proposed in this section has been explained by considering a 1st-order $\Delta\Sigma$ modulator, the same approach is still valid if a higher-order modulator dithers the integer-divider modulus. In that case, only the shape of the time diagrams in Fig. 7 will be different.

As a final remark, we observe that the ADPLL presented in [19] shares some similarities with the present work, since it employs a DTC in the feedback path in place of the TDC. However, the DTC in [19] has 5.3 equivalent bits (i.e. a least-significant bit of 4.7 ps over a DCO period of 189 ps at 5.3 GHz), leading in principle to a quantization cycle in the fractional-N case much larger than the thermal noise. Therefore, although that DTC realizes the one-bit phase comparison as in the proposed case, it is not designed to enforce the random-noise regime, discussed previously.

IV. FRACTIONAL-N DIVIDER DESIGN AND MISMATCH EFFECT CANCELLATION

A. Fractional-N Divider Topology

The practical implementation of the fractional-N divider delay is described in this section. The controllable delay, or DTC, is realized by a switching stage driving MOS capacitors (see Fig. 8): The capacitors can be switched from inversion- to depletion-mode to digitally regulate the delay of the div signal.

For now, let us assume that the block “divider” is a standard integer-N divider, as the one in Fig. 7.

The input full-scale range of the controllable delay, t_g , is nominally set equal to $2T_{dco}$, to ensure that t_g is always larger than T_{dco} over PVT variations. This extra margin would require an 11-bit DTC, to maintain a 10-bit equivalent resolution, as initially requested. The DTC is designed with a segmented-like architecture, as shown in Fig. 8. In order to save area and to reduce the number of control lines, a first bank of 64 identical coarse capacitors is placed in parallel to a second bank of 64 identical fine capacitors, for a total of 12 bits. The unit-size coarse capacitor is nominally 32 times larger than the unit-size fine capacitor, so that the range of the fine bank is nominally twice the least-significant bit (LSB) of the coarse one. This choice let the fine bank cover the LSB of the coarse bank over PVT variations and leads to an 11-bit DTC. Instead of controlling the fine bank by a binary code, as conventionally done in segmented DACs, both capacitor banks are controlled by thermometer codes. This choice reduces the differential nonlinearity (DNL) of the fine bank and adds negligible overhead in terms of number of control lines and decoder area. The resulting time resolution of the fractional-N divider, neglecting for additional PVT margins, is $2^{-10}T_{dco}$. In the present design, the longest period of the DCO is $T_{dco} = 345$ ps (corresponding to 2.9 GHz). Thus, the achievable time resolution is about 340 fs, which is sufficiently fine to enforce the random noise regime.

The two capacitor banks are switched by two 6-bit words, $a_1[k]$ and $a_2[k]$, that are the equivalents of the $a[k]$ signal in the simplified scheme in Fig. 7. The automatic adaptation of the gains, $\hat{g}_1$ and $\hat{g}_2$, which are the counterparts of $\hat{g}$ in Fig. 7, are performed by two LMS algorithms. The latter correlate the 1-bit output of the BB-PD, $e[k]$, with the $q_1[k]$ and $q_2[k]$ signals, respectively.

The block “divider”, in Fig. 8, can divide its input frequency by N or $(N + 0.5)$. This, extra, practical complication arises from the order of the $\Delta\Sigma$ modulator dithering the divider itself. Essentially, simulations show that a 2nd-order modulator is needed, in order to make the LMS algorithm properly converge in the case of a near-integer fractional channel. In this case, the output of a 1st-order $\Delta\Sigma$ modulator would be almost constant. Moreover, the quantization error, i.e. the difference between the $\Delta\Sigma$ modulator output and input, has to be integrated, to provide a $q_1[k]$ signal proportional to the phase error. This signal could display very low-frequency idle tones, while the adaptive filter theory [24] shows that the LMS loop must be much slower than the period of the signal to be cancelled ($q_1[k]$, in this case), in order to operate properly. The output of a 2nd-order $\Delta\Sigma$ modulator, instead, produces a much faster $q_1[k]$ and the LMS loop can be designed to converge quickly.

However, the output of the 2nd-order $\Delta\Sigma$ modulator features 4 levels and it can be shown that the maximum time error at the divider’s output doubles, becoming as wide as $2T_{dco}$, which would require doubling the full-scale delay of the subsequent DTC. This issue is solved thanks to the availability of the two differential DCO outputs, which make it possible to easily obtain a minimum delay equal to $T_{dco}/2$, realizing in practice a fractional divider by $(N + 0.5)$. Fig. 9 shows the scheme of this circuit, a simple phase-interpolator fractional divider, in which

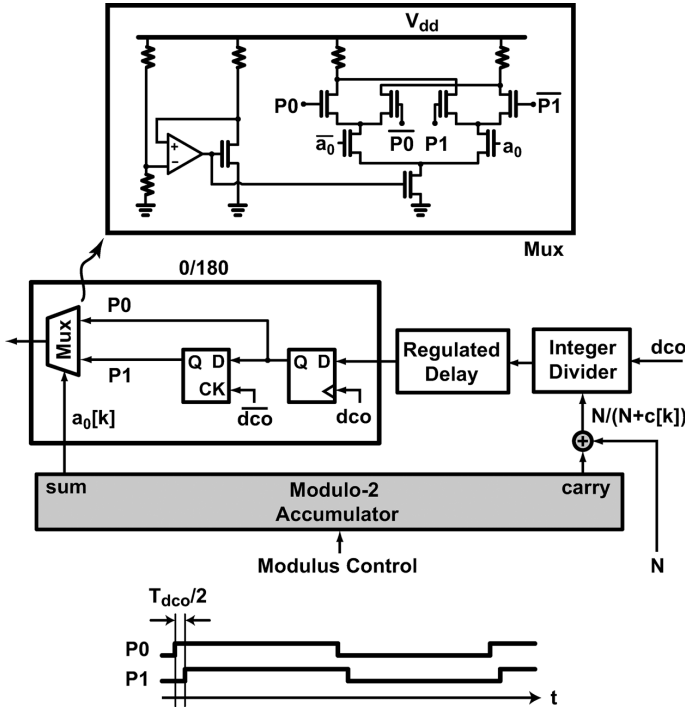


Fig. 9. The $N/(N + 0.5)$ divider along with the multiplexer circuit schematic.

the P0 and P1 signals are obtained by consecutively sampling the output of an integer divider, with the two 0/180 degree outputs of the differential DCO. Before sampling, an automatic control loop adds a regulated delay to the divider output, in order to reduce the probability of metastability. A multiplexer, controlling which one of the two signals is selected, can add a time shift equal to $T_{dco}/2$. The phase interpolator is controlled in the standard way by a modulo- L accumulator ($L = 2$, in this case). In practice, this scheme is analogous to a one-bit delay line. By adopting this solution, the maximum time variation at the divider output with a 2nd-order $\Delta\Sigma$ modulator is limited to T_{dco} .

The other $\Delta\Sigma$ modulator driving the coarse capacitor bank in Fig. 8 is a 1st-order one. Simulations show that, even in the case of a near-integer fractional channel, its input signal is sufficiently dithered and its quantization noise, $q_2[k]$, is obtained without integration, as the phase and not the frequency of the feedback signal is controlled in this case.

B. Correction of Mismatches

Unavoidable mismatches among the components of the fractional divider cause fractional spurs at the PLL output. The most serious problem by far, arises in the $N/(N + 0.5)$ divider just described, since it controls the largest delay. Note that these errors have no impact on the maximum output time step, which is always exactly T_{dco} . The maximum delay is in fact obtained by incrementing the integer division factor from N to $(N + 1)$ and by selecting always the phase P0. The $T_{dco}/2$ delay (obtained by alternatively selecting P0 and P1 while the integer factor toggles between N and $N + 1$) is instead affected by device mismatches and inaccuracies such as a duty cycle different from 50% in the oscillator or mismatches between the propagation delays through the two multiplexer gates. The resulting

effect is essentially the same described in [6], for a phase-interpolator-based divider: In short, let us assume that the division factor is $(N + 0.5)$ and that the delay between P0 and P1 is $(T_{dco}/2 + dt)$. It is easy to derive that the output divider's period, T_{div} , will switch between $[(N + 0.5)T_{dco} + dt]$ and $[(N + 0.5)T_{dco} - dt]$. While the average value of the period is correct, its duration is periodically modulated (peak-to-peak value equal to $2dt$), a fact that gives rise to output spurs.

The digital correlation algorithm in Fig. 10 corrects this impairment while operating in background. When $a_0[k]$ is 1 and the multiplexer selects P1, the error dt is injected into the loop and detected by the BB-PD. Correlating the BB-PD output $e[k]$ with $a_0[k]$, a $\hat{c}$ signal is obtained which is proportional to dt . By multiplying $\hat{c}$ by $a_0[k]$ and subtracting the result from the delay control, as indicated in Fig. 10, dt may be cancelled out.

Note that all the LMS algorithms, sketched both in Fig. 8 and Fig. 10, exploit the output $e[k]$ of the BB-PD, which is a single-bit signal. This simplifies the digital correlators, since the multiplications reduce to simple changes of sign, resulting in a reduction of area and power consumption. In the latter case discussed, the algorithm is further simplified since $a_0[k]$ is also a single-bit signal, making straightforward the multiplication between $\hat{c}$ and $a_0[k]$.

V. CIRCUIT IMPLEMENTATION

The adoption of the bang-bang architecture allows focusing the design effort on few critical circuits, namely the 0/180 phase shifter in Fig. 9 and the D flip-flop that implements the BB-PD. Current-mode logic (CML) has been used for these building blocks, since it guarantees better immunity to disturbances and noise, it strongly reduces the injection of disturbances in the power supply and, in general, it helps in eliminating other mechanisms of generation of fractional spur. As an example of this approach, Fig. 9 sketches the schematic of the multiplexer used in the fractional divider, in which a simple servo circuit regulates the voltage drop across the load resistors. With respect to a CMOS-logic implementation, CML logic requires more power consumption but, given the limited number blocks in which it has been used, the dissipation excess is negligible. By contrast, CML, given its constant current biasing, allows reducing the amount of supply decoupling capacitance.

The simplified scheme of the PLL forward path, following the 1-bit phase detector, is sketched in Fig. 11. It includes the proportional-integral digital filter, whose output is interpolated by a first-order $\Delta\Sigma$ modulator. The 5-bit output of the $\Delta\Sigma$ modulator is fed to a voltage DAC, realized by a resistors string. After the RC filtering, the output of the voltage DAC drives the node of an analog varactor, covering a frequency range of about 3 MHz. Multiple tuning characteristics, covering the entire oscillator tuning range (from 2.9 to 4.0 GHz), are obtained by means of three scaled banks of capacitors within the tank. The overlap between the characteristics is such that, when the system is in lock, the tuning voltage at the output of the voltage DAC always exploits the central portion of the tuning characteristic (roughly ± 750 kHz from the center frequency). When the DAC exceeds this range, one of the finer capacitors is switched on or off accordingly. In this way, the PLL operates as close as possible to the center of one tuning curve and it maintains

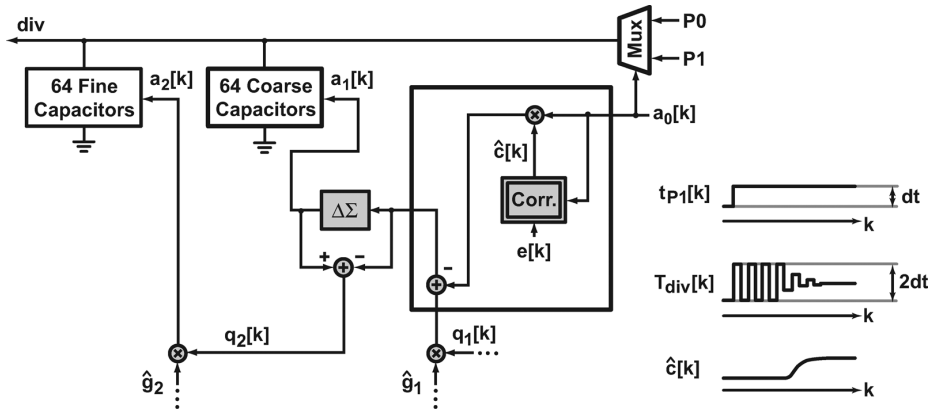


Fig. 10. Digital background cancellation of the time error at the output of the $N/(N + 0.5)$ divider.

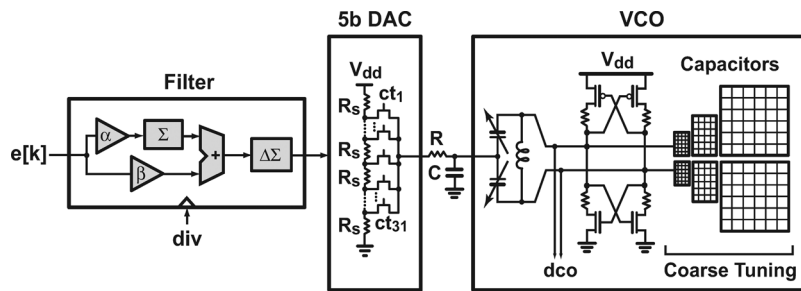


Fig. 11. Simplified circuit schematic of the forward path.

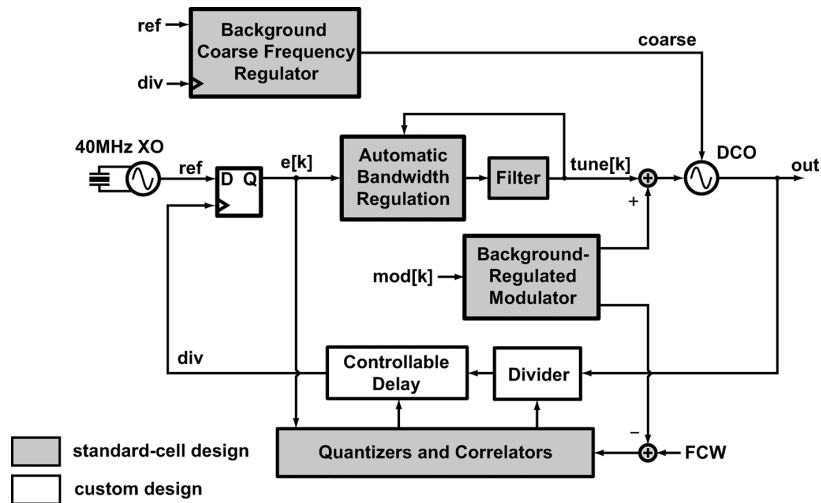


Fig. 12. Block schematic of the entire synthesizer.

lock even in the presence of small temperature or voltage drifts. Tracking of larger perturbations is performed by the frequency control loop, which sets the control words of the capacitor DACs and operates continuously. The latter loop will be briefly described in the following.

The topology of the transconductor employed in the oscillator is also shown in Fig. 11. In a low-noise low-power design, great care must be devoted to the design of this component. In particular, flicker (or $1/f$) noise can be very high in scaled CMOS technologies and the resulting $1/f^3$ phase noise can dominate over the $1/f^2$ component up to frequencies that exceed the PLL bandwidth, unless the mechanisms responsible for the flicker noise

upconversion into $1/f^3$ phase noise are carefully addressed. To the purpose of avoiding one source of flicker noise, the tail generator has been removed. Moreover, a resistor is added in series to each drain of the switching transistors. This solution reduces the upconversion of the residual flicker noise of the cross-coupled pairs, as discussed in [25].

Fig. 12 shows the schematic of the entire synthesizer, the standard-cell-based digital blocks are filled in gray. Besides the circuits previously described, three other implemented blocks are sketched in this figure, respectively a coarse frequency regulator, an automatic bandwidth regulation and a direct frequency modulator. The first block is needed to speed up the lock tran-

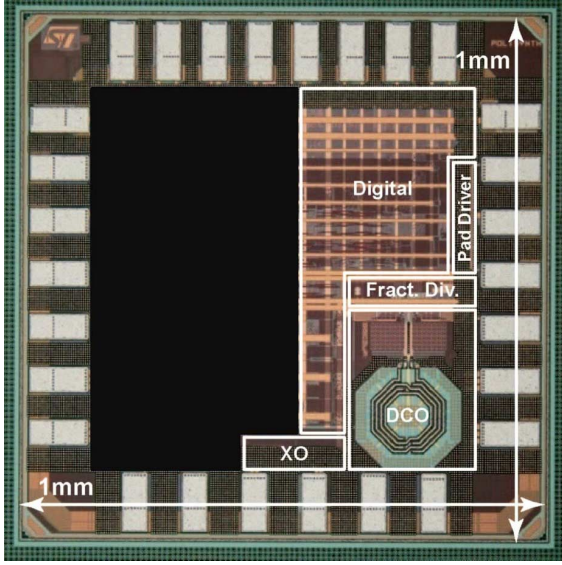


Fig. 13. Chip photograph (only a portion of the die area is occupied by the synthesizer).

sient, that otherwise would result too slow given the very narrow linear input range of the BB-PD [21]. In this design, this range results about 1 ps wide, which would be inadequate for recovering the lock in less than 1 ms, in the case of large frequency steps. To solve this issue, the coarse frequency-control loop sets the control words of the three capacitor banks within the DCO tank. This digital regulator does not rely on a state machine working only at the PLL startup, but it is an all-digital loop always operating in background. The only difference with respect to the main PLL loop is that it applies no variations to the tuning control words, until the ref and div signals are separated by about ± 100 ps. This method allows fast tracking of large temperature or voltage perturbations, which should drive the DCO tuning out of its permitted range.

The digital circuit, performing the automatic regulation of the loop bandwidth, avoids the undesirable dependence of the bandwidth itself on the input jitter (through the BB-PD linearized gain). It is alike to the automatic bandwidth regulator described in [26]. In contrast to that implementation, the quantization noise of the $\Delta\Sigma$ modulator driving the DCO is used to normalize the loop gain and no additional pseudo-random sequence is needed. Finally, a two-point frequency modulator is also implemented, essentially to demonstrate the capability of the synthesizer as a wideband frequency modulator. The automatic background regulation of the loop parameters is used even in the modulator mode.

VI. EXPERIMENTAL RESULTS

Fig. 13 shows the die photograph of the prototype chip, fabricated in a 65-nm CMOS technology. The synthesizer occupies an active area of about 0.22 mm^2 . Either an external reference or an on-chip crystal oscillator can be used to provide the reference signal to the test chip. In the following measurements, only the internal oscillator has been used as a reference and its measured frequency is 40.0098 MHz. The total power consumption (excluding the pad drivers and the crystal oscillator) is 4.5 mW,

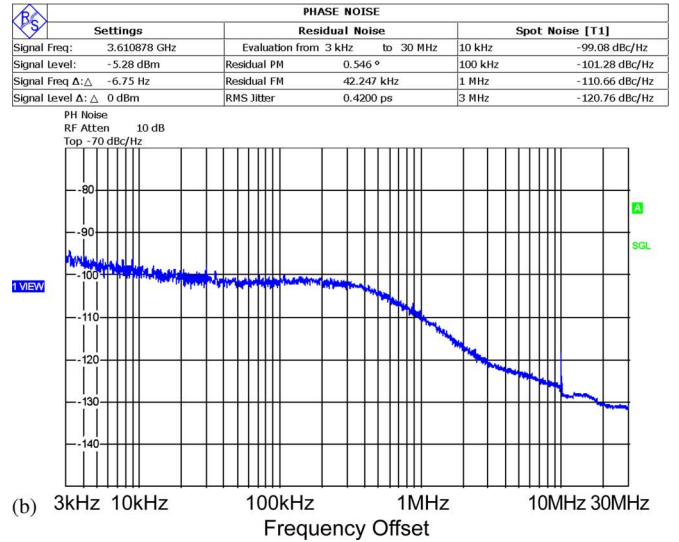
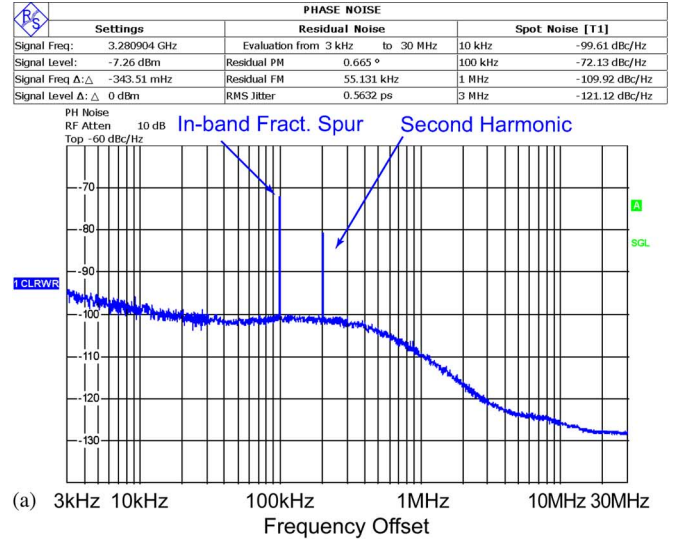


Fig. 14. Measured phase-noise spectra: (a) channel with in-band fractional spurs and (b) channel with no in-band spurs.

from a 1.2-V supply, which is mainly dissipated by the feedback path (49%), i.e. divider plus controllable delay, and by the DCO (38%).

A. Spectral Purity and Figure-of-Merit

In addition to the power dissipation, the other figure entering in the expression of the FoM is the jitter, which is proportional to the integral of the phase-noise spectrum. Fig. 14 shows two spectra obtained for different synthesized channels, the in-band noise plateau is about -102 dBc/Hz at 50 kHz in both cases and it does not appreciably change over the entire PLL tuning range. According to simulations, the main contributors to this plateau are the thermal noise of the delay stage and the BB-PD and the folded quantization noise arising from the DCO frequency granularity [27]. In all the measurements, the bandwidth is automatically set to 312 kHz (i.e. $f_{\text{ref}}/128$), which is the value that optimizes the random noise component of the jitter, i.e. the jitter evaluated without taking into account the presence of spurious tones. Clearly, the evaluation of the jitter is

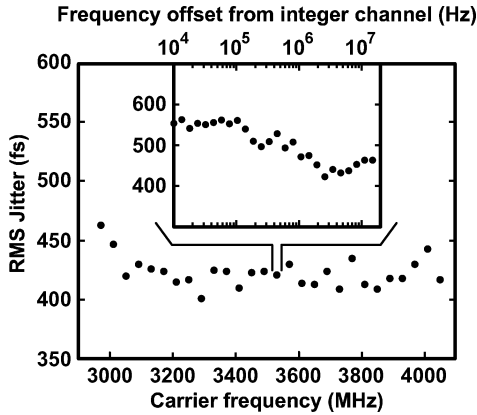


Fig. 15. Measured RMS jitter over the synthesizer tuning range.

significant when performed for channels whose fractional frequency falls within the PLL bandwidth. In fact, the worst-case jitter is obtained from the spectrum shown in Fig. 14(a), measured when the fractional channel occurs at 100 kHz from an integer one. A -42 -dBc fractional spur, the highest measured, at 100-kHz offset and its second harmonic occur within the PLL bandwidth.³ Most likely, these residual tones are due to a specific behavior of the DTC embedded in the feedback path. In fact, the delay added to the feedback signal increases more than linearly, with respect to the number of load capacitors. This systematic nonlinearity gives rise to an integral nonlinearity (INL), responsible for the residual spur level. The RMS jitter is about 560 fs, as obtained from the phase-noise analysis software of the spectrum analyzer. This software integrates the spectrum in the visualized frequency range. In this case, the integration range spans from 3 kHz to 30 MHz. However, no difference in jitter is observed extending this range down to 1 kHz or up to 100 MHz, since the dominant components of the spectrum are those in the 100-kHz-to-1-MHz frequency range. This figure decreases, when evaluated with channels featuring a fractional frequency much larger than the loop bandwidth. For instance, Fig. 14(b) shows a case in which the fractional spur occurs at 10-MHz offset, where it is attenuated by the loop filter. In this case, the resulting RMS jitter is about 420 fs. Measurements have been also repeated, with no off-chip regulation of loop parameters, over the fractional synthesized channels from 2962 to 4042 MHz in steps of 40 MHz and from 3520 MHz (integer channel) to 3540 MHz with logarithmic frequency spacing. The resulting RMS jitter as a function of the synthesized frequency is reported in Fig. 15. The FoM, as defined in the introduction, that expresses the jitter-power trade-off is -238.3 dB, in the worst case, and it improves to -241.3 dB, when fractional spurs fall out of the PLL bandwidth.

The spectral purity at large offsets has been also measured, although its contribution to the total jitter is negligible. Concerning the reference spur, it results below -72 dBc over the entire tuning range. The spectrum in Fig. 14, which was obtained from the phase-noise analysis software of the spectrum analyzer, has a noise floor limited to about -130 dBc/Hz and it does not

³The phase-noise analysis software visualizes a lower spur level (-72 dBc), since it divides the spur power by its resolution bandwidth. The correct value of -42 dBc is either obtained by correcting for this mechanism or by the direct measurement presented in Fig. 17(b).

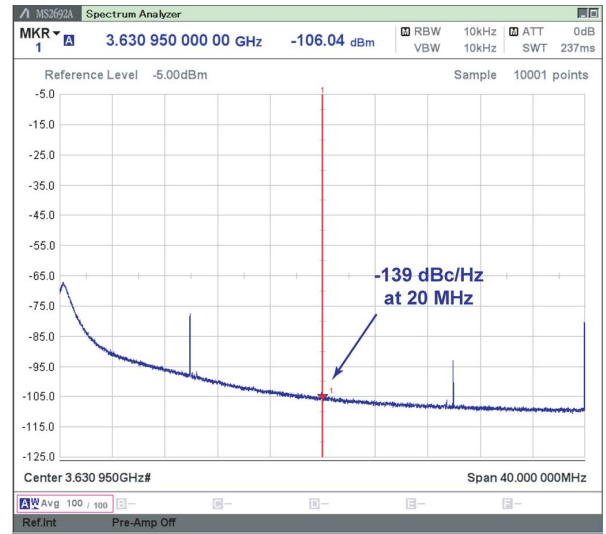


Fig. 16. Measured out-of-band phase noise.

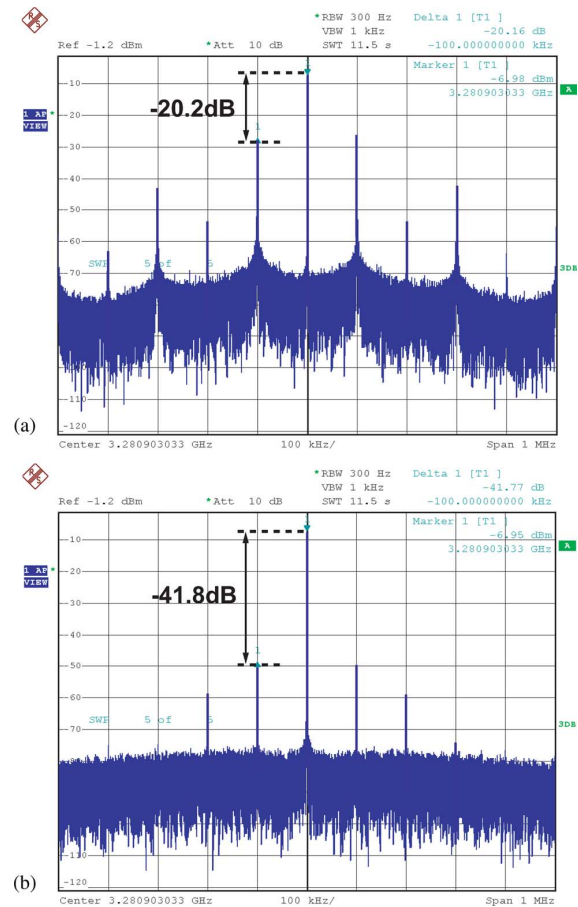


Fig. 17. Output spectrum before (a) and after (b) the enabling of the background calibration algorithms.

allow measuring the PLL out-of-band noise. The phase noise at large offset, obtained without relying on the phase-noise analysis software, is shown in Fig. 16 for a carrier power of -7 dBm. The measured phase noise at 20 MHz offset is -139 dBc/Hz, which can be ascribed to the $1/f^2$ component of the DCO phase noise.

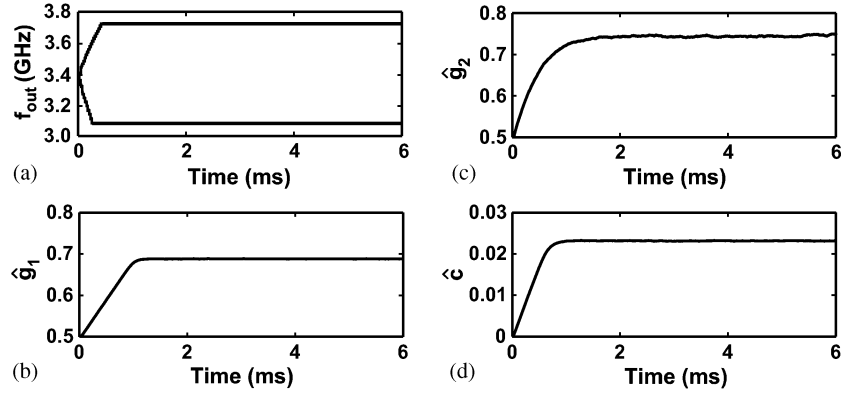


Fig. 18. Measured transients of the coarse frequency regulator and of the LMS coefficients.

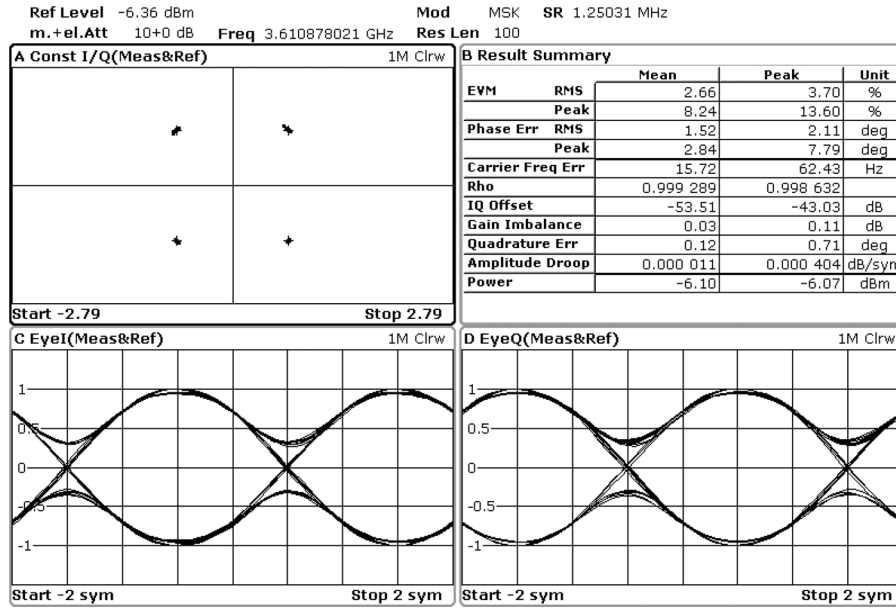


Fig. 19. Measurements of the GMSK frequency modulator.

B. Background Calibration Algorithms and Modulation Capability

The impact of the background calibration algorithms, implemented in the circuit, can be appreciated with the help of the measurements reported in Fig. 17. They show the PLL output spectrum when a fractional channel is synthesized respectively before, (a), and after, (b), enabling of the delay-stage gain regulation and the mismatch correctors (i.e. the coefficients $\hat{g}_1$, $\hat{g}_2$ and $\hat{c}$). The frequency span is 1 MHz and the channels are at 100-kHz offset from the integer ones, so that the fractional spur falls within the PLL bandwidth. The much larger noise in the first spectrum arises from the inaccurate cancellation of the $\Delta\Sigma$ quantization error and from the BB-PD hard nonlinearity. This noise causes a decrease of the BB-PD equivalent gain and, consequently, of the loop bandwidth. In that case, the RMS jitter is so large, 5.56 ps, that it exceeds the dynamic range of the automatic bandwidth regulation system. When the calibration algorithms are enabled, the fractional spur falls down to a level of -42 -dBc and the loop bandwidth is regulated correctly.

The coarse frequency locking behavior, in the presence of large frequency steps, is shown in Fig. 18(a). Two measurements

are presented, in both cases starting from 3.4 GHz and featuring a 350-MHz frequency step, but with opposite signs. Thanks to the coarse frequency regulator, the locking is achieved in about 400 μ s. Those measurements have been obtained by down-converting the PLL output signal to baseband, then sampling it with a digital scope. The noise associated to the measured frequency waveform is due to the limited sampling frequency (2 GS/s) of the scope.

The test chip allows also measuring the transient values of the coefficients $\hat{g}_1$, $\hat{g}_2$ and $\hat{c}$. The measured transients from startup to the final value are shown in Fig. 18(b), (c), (d). They all have settled after 700 μ s. In the case of small perturbations, although the LMS loops are nonlinear, they can be linearized (see [28] for an intuitive derivation). Under this assumption, the loop time constant in the presented design is about 4.5 μ s, which is fast enough to track temperature and voltage supply drifts.

Although a single-bit PD is adopted, the PLL still allows the two-point modulation scheme for the direct frequency modulation. This scheme enlarges the modulation bandwidth of the PLL over the limit of the loop bandwidth, which can be instead optimized for jitter performance. To demonstrate this capability in the bang-bang case, a GMSK modulation at 1.25-Mb/s bit

TABLE I
PERFORMANCE SUMMARY COMPARED WITH OTHER LOW-JITTER FRACTIONAL-N PLLS

	[5]	[29]	[4]	[30]	This work
Architecture	Digital	Digital	Digital	Analog	Digital
Reference (MHz)	N/A	40	50	35	40
Output (GHz)	5-12	9.2-12	3.67	2.1	2.9-4.0
Tuning Range	82.3%	26.4%	N/A	N/A	31.9%
RMS Jitter (fs) min-max	560* - N/A	597* - N/A	300* - N/A	468 - N/A	400* - 560
Bandwidth (kHz)	500	500	500	700	312
Out-of-band Fract. (dBc)	-48 (2MHz)	N/A	-53 (1MHz)	N/A	-53 (1MHz)
In-band Fract. (dBc)	N/A	N/A	-42	-60	-42
Reference Spur (dBc)	-56	N/A	-65	N/A	-72
Phase noise at 20MHz (dBc/Hz)	-145	-140	-150	-150	-139
Power Diss. (mW)	30	50	47	28	4.5
Area occup. (mm ²)	0.28	1.2	0.9	N/A	0.22
Process (nm)	40	90	130	180	65

* integrated jitter without in-band fractional spurs

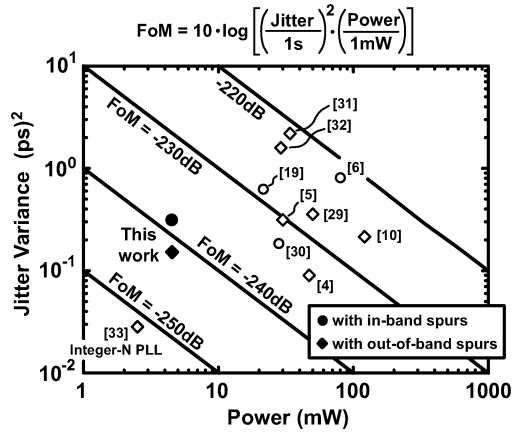


Fig. 20. Jitter variance vs. power consumption for recently published fractional-N synthesizers.

rate entailing a frequency step of 312 kHz has been applied to the PLL. The measured error vector magnitude (EVM) of the demodulated signal is only 2.66% or, equivalently, -31.5 dB. Fig. 19 shows the constellation of the received data, the summary of the measurement results and the eye diagrams of the I and Q baseband signals, as obtained from the signal analyzer.

C. Performance Summary and FoMs Comparison

The main features of the present circuit are summarized in Table I, where they are compared with other published fractional-N PLLs providing the measured absolute jitter performance. Fig. 20 compares the FoM of this PLL with the same figure evaluated for other fractional-N PLLs, including the bang-bang ADPLLs in [19]. In some cases (circles), the in-band spurs are accounted in the jitter evaluation. In other cases (diamonds), they are not. The -251 -dB FoM of the integer-N synthesizer presented in [3] is also reported, since it represents the best result available in open literature. Despite a large FoM gap still exists between this work and the best integer-N PLL, the presented synthesizer shows the best jitter-power trade-off

among the fractional-N PLLs, still preserving all the features of a synthesizer for wireless applications.

VII. CONCLUSIONS

A low-power low-jitter $\Delta\Sigma$ fractional-N digital frequency synthesizer has been realized thanks to the substitution of the power-hungry multi-bit TDC with a bang-bang, or single-bit, TDC. Detrimental effects, such as limit cycles, caused by this hard nonlinearity within the loop, are avoided by the introduction of a fractional-N divider with 10 b equivalent resolution, and the use of digital background calibration algorithms. The synthesizer dissipates 4.5 mW, with a worst-case jitter of $560 f_{s_{rms}}$ (FoM of -238.3 dB), which reduces to $420 f_{s_{rms}}$ (FoM of -241.3 dB) when fractional spurs are not present. The prototype also includes a two-point modulation scheme, which demonstrates a 1.25-Mb/s GMSK frequency modulation with -31.5 -dB EVM.

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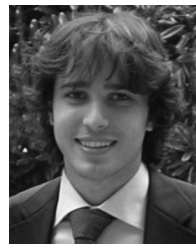
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